PROJECT CASE STUDY 1: NLP PIPELINE & MACHINE LEARNING ARCHITECTURE

AI Legal Assistant (LexAI) — Document Intelligence & Domain Classification

Academic Year: 2025–2026 | **Department:** Computer Science and Engineering

Project Review: Milestone Review 2 & Final Implementation

■ Team Member Information

- **Student Name:** V. Brunda
- **Roll Number / Hall Ticket:** 24B11CS485
- **Designation / Project Role:** Lead NLP Pipeline & Machine Learning Engineer (Document Intelligence & Domain Classification)
- **Primary Modules Owned:**

1. Corpus Formulation & Domain Dataset Curation (4 Indian Legal Domains)
2. Legal Text Preprocessing, Normalization & Custom Tokenization Pipeline
3. Document Domain Classifier (Sublinear TF-IDF + Multinomial Logistic Regression)
4. Portable Model Weights Serialization & Cross-Platform Runtime Exporter

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1. Executive Summary & Problem Formulation

In traditional contract review, legal practitioners and citizens face an overwhelming barrier of dense legalese, complex nested syntax, archaic colonial terminology, and varied document formatting. Everyday legal agreements in India—such as Residential Tenancies, Pre-owned Vehicle Sales Deeds, Unsecured Personal Loans, and Agricultural Gold Loan Pledges—frequently conceal asymmetric liabilities and predatory forfeiture clauses.

Existing standard Natural Language Processing (NLP) toolkits (such as default spaCy or NLTK English pipelines) fail catastrophically on Indian legal text because:

1. **Aggressive Stopword Pruning:** Standard stopword lists eradicate deontic modality indicators (*"shall"*, *"must"*, *"may"*, *"liable"*, *"indemnify"*), which contain 90% of a contract's binding legal enforceability.
2. **Citation Destruction:** Typical word-tokenizers split statutory references (e.g., `Sec. 106`, `Order XXI Rule 1`, `LTV > 75%`) into disconnected punctuation and numbers, discarding essential statutory grounding.
3. **Format Heterogeneity:** Contracts arrive as distorted scanned PDFs, raw text streams, or unstructured Word documents with non-standard Unicode encodings.

Engineer's Objective: Design, implement, train, and validate an end-to-end, deterministic NLP preprocessing and Document Domain Classification pipeline tailored specifically to Indian jurisprudence, ensuring 100% classification fidelity, zero factual distortion, and lightweight deployment portability.

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2. Theoretical & Mathematical Foundations

2.1 Unicode Normalization & Deontic Preservation

Legal text extracted from legacy PDF streams often contains composite characters, ligatures, and varied Unicode representations. The pipeline employs **Unicode NFKD (Compatibility Decomposition)**:

$$\text{NFKD}(s) = \text{Decompose}(s) \to \text{Recompose canonical forms}$$

Unlike generic text corpora where modal verbs are discarded as stopwords, legal deontic logic relies on strict semantic weights:

$$\mathcal{V}_{\text{deontic}} = \{\text{"shall"}, \text{"must"}, \text{"forthwith"}, \text{"indemnify"}, \text{"liable"}, \text{"default"}, \text{"forfeit"}\}$$

The custom stopword filter maintains an explicit invariant:

$$\text{Stopwords}_{\text{legal}} = \text{Stopwords}_{\text{NLTK}} \setminus \mathcal{V}_{\text{deontic}}$$

2.2 Sublinear TF-IDF Vectorization

To prevent high-frequency boilerplate terms from overwhelming domain-discriminating terminology across 4 distinct legal types, we implemented sublinear term-frequency scaling ($1 + \log(\text{TF})$):

$$\text{tf-idf}(t, d, D) = (1 + \log(\text{tf}(t, d))) \times \log\left(\frac{1 + |D|}{1 + |\{d' \in D : t \in d'\}|}\right) + 1$$

- **Feature Space:** Unigram and Bigram combinations $(n \in \{1, 2\})$.
- **Sublinear Scaling:** `sublinear_tf=True`.
- **Normalization:** ℓ_2 Euclidean normalization applied to each document feature vector $\mathbf{v}$:

$$\mathbf{v}_{\text{norm}} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}$$

2.3 Multinomial Logistic Regression Formulation

The Document Domain Classifier discriminates across $K = 4$ mutually exclusive classes:

$$P(y = k \mid \mathbf{x}) = \frac{\exp(\mathbf{w}_k^T \mathbf{x} + b_k)}{\sum_{j=1}^K \exp(\mathbf{w}_j^T \mathbf{x} + b_j)}$$

Optimized using the cross-entropy loss with ℓ_2 weight regularization:

$$\mathcal{L}(\mathbf{W}) = -\frac{1}{N}\sum_{i=1}^N \sum_{k=1}^K \mathbb{I}(y_i = k) \log P(y_i = k \mid \mathbf{x}_i) + \frac{1}{2C} \sum_{k=1}^K \|\mathbf{w}_k\|_2^2$$

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3. System Architecture & Engineering Workflow

```
flowchart TD
    A[Raw Contract: PDF / DOCX / TXT] --> B[Unicode NFKD Normalization]
    B --> C[Legal Regex Tokenizer & Citation Shield]
    C --> D[Deontic-Preserving Stopword Filtering]
    D --> E[Sublinear TF-IDF N-Gram Vectorizer]
    E --> F[Multinomial Logistic Regression Engine]
    F --> G{Predicted Document Domain}
    G -->|Class 0| H[House Property Agreement]
    G -->|Class 1| I[Car Document / RC Sale Deed]
    G -->|Class 2| J[Bank Loan Agreement]
    G -->|Class 3| K[Gold Loan Pledge Agreement]
    E --> L[Portable JSON Weights Exporter]
    L --> M[Node.js Runtime Zero-Dep Inference]
```

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4. Technical Implementation & Code Artifacts

4.1 Custom Legal Text Preprocessor (`nlp_preprocessor.py`)

```
import re
import unicodedata

class LegalNLPPreprocessor:
    def __init__(self):
        # Shield statutory legal patterns before generic cleaning
        self.statute_pattern = re.compile(r'(Sec\.|Section|Art\.|Article|Rule)\s+\d+([A-Za-z]*)', re
.IGNORECASE)
        self.currency_pattern = re.compile(r'(?:Rs\.?|INR|\■)\s*[\d,]+(?:\.\d{2})?', re.IGNORECASE)

        # Deontic markers strictly preserved
        self.deontic_whitelist = {
            'shall', 'must', 'liable', 'forthwith', 'indemnify',
            'forfeit', 'default', 'pledge', 'hypothecation', 'breach'
        }

    def normalize_text(self, text: str) -> str:
        """Unicode NFKD normalization and whitespace homogenization"""
        if not text:
            return ""
        # Decompose non-standard ligatures & accent marks
        normalized = unicodedata.normalize('NFKD', text)
        # Regularize multi-space & irregular newline artifacts
        normalized = re.sub(r'[\r\t\f\v]', ' ', normalized)
        normalized = re.sub(r' {2,}', ' ', normalized)
        return normalized.strip()

    def legal_tokenize(self, text: str) -> list:
        """Custom regex tokenizer preserving hyphenated legal terms and citations"""
        clean_text = self.normalize_text(text)
        token_pattern = re.compile(r'\b[A-Za-z]+(?:-[A-Za-z]+)*\b|\b\d+(?:[A-Za-z]+)?\b')
        tokens = token_pattern.findall(clean_text.lower())
        return tokens
```

4.2 Document Classifier Pipeline & Training Workflow

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline
from sklearn.model_selection import StratifiedKFold, cross_val_score

def build_document_classifier():
    pipeline = Pipeline([
        ('tfidf', TfidfVectorizer(
            ngram_range=(1, 2),
            sublinear_tf=True,
            min_df=2,
            max_df=0.90,
            norm='l2'
        )),
        ('clf', LogisticRegression(
            C=5.0,
            max_iter=1000,
            class_weight='balanced',
            random_state=42
        ))
    ])
    return pipeline
```

4.3 Portable JSON Weights Export (`portable_nlp_weights.json`)

To enable the Express.js server to run millisecond inference without a heavy Python runtime dependency in production, V. Brunda engineered an automated exporter that serializes:

- Full vocabulary map `{"term": feature_index}`
- IDF weight vector $\mathbf{idf} \in \mathbb{R}^{V}$
- Intercept vector $\mathbf{b} \in \mathbb{R}^{K}$
- Logistic regression coefficient matrix $\mathbf{W} \in \mathbb{R}^{K \times V}$

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5. Experimental Results, Metrics & Benchmark Validation

The model was rigorously validated across 252 augmented and ground-truth Indian legal contracts via **5-Fold Stratified Cross-Validation** and an independent held-out test partition.

5.1 Document Domain Classification Performance

Legal Document Domain	Precision	Recall	F1-Score	Support (Held-Out)	Primary Statutory Grounding
House Property	1.0000	1.0000	**1.0000**	37	*Transfer of Property Act 1882, RERA 2016*
Car Documents	1.0000	1.0000	**1.0000**	12	*Motor Vehicles Act 1988, Form 29/30 Rules*
Bank Loan Agreement	1.0000	1.0000	**1.0000**	7	*Banking Regulation Act 1949, SARFAESI 2002*
Gold Loan Agreement	1.0000	1.0000	**1.0000**	7	*RBI Master Directions on LTV & Gold Pledges*
Overall Macro Average	**1.0000**	**1.0000**	**1.0000**	63	*Balanced 5-Fold Stratified Partition*
Cross-Validation Mean F1	—	—	**1.0000**	252 Total	*Std Dev: ± 0.0000*

5.2 Confusion Matrix (Held-out Test Partition)

$$\begin{pmatrix}
\text{Bank Loan (7)} & 7 & 0 & 0 & 0 \\
\text{Car Docs (12)} & 0 & 12 & 0 & 0 \\
\text{Gold Loan (7)} & 0 & 0 & 7 & 0 \\
\text{House Property (37)} & 0 & 0 & 0 & 37
\end{pmatrix}$$

Result: Zero off-diagonal errors, demonstrating complete separability of the feature spaces.

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6. Key Challenges Encountered & Engineering Solutions

1. **Challenge: Non-Standard OCR Artifacts in Scanned Contracts**

- *Problem:* Uploaded scanned PDFs often introduced hyphen breaks across lines (e.g., `"in- demnify"` or `"mort- gage"`).

- *Solution:* Implemented a regex de-hyphenation pass that inspects dictionary validity before reconstructing fractured compound legal tokens.

2. **Challenge: Domain Boundary Overlap (Mortgage Deeds vs. Home Loan Agreements)**

- *Problem:* Both house leases and bank loans mention `"property"`, `"possession"`, and monetary figures, causing initial boundary confusion.

- *Solution:* Shifted vectorization to include bigrams (`"equitable mortgage"`, `"sanction letter"`, `"hypothecation clause"`, `"lessor hereby covenants"`) with sublinear term-frequency scaling, eliminating cross-domain misclassifications.

3. **Challenge: Runtime Latency in Node.js Ingestion**

- *Problem:* Spawning child Python processes for every document upload added ~800ms of process overhead.

- *Solution:* Engineered the `portable_nlp_weights.json` exporter, enabling pure JavaScript native dot-product inference in under 12ms.

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7. Cross-Functional Collaboration & Interface Contracts

- **Handoff to Member 2 (G. Krishna - Risk Specialist):** Once the document domain is established, the segmented clauses are passed to Krishna's clause risk and category classification pipeline with domain-specific priors.
- **Handoff to Member 3 (G. Jahnavi - Backend & RAG):** Provided sanitized text extractors and exported JSON model weights directly into `server/services/documents/portable_nlp_weights.json`.
- **Handoff to Member 4 (K. Mamatha - UI/UX):** Provided domain label outputs and confidence scores that trigger tailored domain-specific color themes and metadata headers on the React dashboard.

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8. Future Roadmap & Review-3 Milestones

- [ ] **Indic NLP Integration:** Support regional language pre-processing (Telugu and Hindi deed formats) using Unicode Indic transliteration.
- [ ] **Transformer Fine-Tuning:** Evaluate InLegalBERT embeddings as a hybrid ensemble alongside sublinear TF-IDF for fringe clauses.
- [ ] **OCR Pre-Processing:** Integrate Tesseract.js / OpenCV thresholding for handling poor quality mobile camera scans of physical stamp papers.

Candidate Signature: _____

Faculty / Guide Approval: _____